



GSFA Online

Updated up to the 48th Session of the Codex Alimentarius Commission (2025)

FOOD ADDITIVE DETAILS

Carbon dioxide (290)

Synonym(s)

- Carbonic Acid Anhydride
- Dry Ice (Solid Form)

Functional Classes

- Carbonating agent
- Foaming agent
- Packaging gas
- Preservative
- Propellant

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CSV (Tables)

GSFA Provisions for Carbon dioxide

Number	Food Category	Max Level	Notes
14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	GMP	Note 59 Note 160
13.2	Complementary foods for infants and young children	GMP	Note 59
14.1.2.3	Concentrates for fruit juice	GMP	Note 69 Note 127
14.1.3.3	Concentrates for fruit nectar	GMP	Note 69 Note 127
01.2.1.2	Fermented milks (plain), heat-treated after fermentation	GMP	Note 59 Note 628
01.2.1.1	Fermented milks (plain), not heat-treated after fermentation	GMP	Note 628
13.1.2	Follow-up formulae	GMP	Note 59
13.1.3	Formulae for special medical purposes for infants	GMP	Note 59
06.4.1	Fresh pastas and noodles and like products	GMP	Note 59 Note 211
14.1.2.1	Fruit juice	GMP	Note 69
14.1.3.1	Fruit nectar	GMP	Note 69
14.2.3	Grape wines	GMP	Note 60
13.1.1	Infant formulae	GMP	Note 59
14.1.1.1	Natural mineral waters and source waters	GMP	Note 466
04.1.1.3	Peeled or cut fresh fruit	GMP	Note 59
01.2.2	Rennet milk (plain)	GMP	Note 59
09.2.5	Smoked, dried, fermented, and/or salted fish and fish products, including mollusks, crustaceans, and echinoderms	GMP	Note 59 Note XS244 Note XS222 Note 382 Note XS167 Note XS189 Note XS236

Number	Food Category	Max Level	Notes
 01.4.2	Sterilized and UHT creams, whipping and whipped creams, and reduced fat creams (plain)		GMP  Note 278  Note 59
 14.1.1.2	Table waters and soda waters		GMP  Note 466

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

GSFA Table 3 Provisions

Carbon dioxide is a food additive that is included in **Table 3**, and as such may be used in the following foods under the conditions of good manufacturing practices (GMP) as outlined in the Preamble of the Codex GSFA. Although not listed below, Carbon dioxide could also be used in heat-treated butter milk of food category 01.1.1 and spices of food category 12.2.1. Note that food categories listed in the **Annex to Table 3** were excluded accordingly. **Carbon dioxide** is acceptable in foods conforming to the following commodity standards: CS 221-2001 (for whipped products only), CS 275-1973, CG 95-2022.

Carbon dioxide is a: Packaging gas, Preservative

Any Packaging gas listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007.

Any Preservative listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 291-2010, CS 256-2007, CS 306-2011.

Number	Food Category
 01.1.4	Flavoured fluid milk drinks
 01.3	Condensed milk and analogues (plain)
 01.4.3	Clotted cream (plain)
 01.4.4	Cream analogues
 01.5	Milk powder and cream powder and powder analogues (plain)
 01.6.1	Unripened cheese
 01.6.2	Ripened cheese
 01.6.4	Processed cheese
 01.6.5	Cheese analogues
 01.7	Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt)
 01.8.1	Liquid whey and whey products, excluding whey cheeses
 02.2.2	Fat spreads, dairy fat spreads and blended spreads
 02.3	Fat emulsions mainly of type oil-in-water, including mixed and/or flavoured products based on fat emulsions
 02.4	Fat-based desserts excluding dairy-based dessert products of food category 01.7
 03.0	Edible ices, including sherbet and sorbet
 04.1.2	Processed fruit
 04.2.2.2	Dried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds
 04.2.2.3	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds in vinegar, oil, brine, or soybean sauce
 04.2.2.4	Canned or bottled (pasteurized) or retort pouch vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds
 04.2.2.5	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g., peanut butter)
 04.2.2.6	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed pulps and preparations (e.g. vegetable desserts and sauces, candied vegetables) other than food category 04.2.2.5
 04.2.2.8	Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds
 05.0	Confectionery
 06.3	Breakfast cereals, including rolled oats
 06.4.3	Pre-cooked pastas and noodles and like products
 06.5	Cereal and starch based desserts (e.g. rice pudding, tapioca pudding)
 06.6	Batters (e.g. for breading or batters for fish or poultry)
 06.7	Pre-cooked or processed rice products, including rice cakes (Oriental type only)

Number ▼	Food Category ▼
 06.8	Soybean products (excluding soybean-based seasonings and condiments of food category 12.9)
 07.0	Bakery wares
 08.2	Processed meat, poultry, and game products in whole pieces or cuts
 08.3	Processed comminuted meat, poultry, and game products
 08.4	Edible casings (e.g. sausage casings)
 09.3	Semi-preserved fish and fish products, including mollusks, crustaceans, and echinoderms
 09.4	Fully preserved, including canned or fermented fish and fish products, including mollusks, crustaceans, and echinoderms
 10.2.3	Dried and/or heat coagulated egg products
 10.3	Preserved eggs, including alkaline, salted, and canned eggs
 10.4	Egg-based desserts (e.g. custard)
 11.6	Table-top sweeteners, including those containing high-intensity sweeteners
 12.2.2	Seasonings and condiments
 12.3	Vinegars
 12.4	Mustards
 12.5	Soups and broths
 12.6	Sauces and like products
 12.7	Salads (e.g. macaroni salad, potato salad) and sandwich spreads excluding cocoa- and nut-based spreads of food categories 04.2.2.5 and 05.1.3
 12.8	Yeast and like products
 12.9	Soybean-based seasonings and condiments
 12.10	Protein products other than from soybeans
 13.3	Dietetic foods intended for special medical purposes (excluding products of food category 13.1)
 13.4	Dietetic formulae for slimming purposes and weight reduction
 13.5	Dietetic foods (e.g. supplementary foods for dietary use) excluding products of food categories 13.1 - 13.4 and 13.6
 13.6	Food supplements
 14.1.4	Water-based flavoured drinks, including "sport," "energy," or "electrolyte" drinks and particulated drinks
 14.2.1	Beer and malt beverages
 14.2.2	Cider and perry
 14.2.4	Wines (other than grape)
 14.2.5	Mead
 14.2.6	Distilled spirituous beverages containing more than 15% alcohol
 14.2.7	Aromatized alcoholic beverages (e.g. beer, wine and spirituous cooler-type beverages, low alcoholic refreshers)
 15.0	Ready-to-eat savouries
 16.0	Prepared foods

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).



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